WORKED SOLUTION

STEP II 2011, Question 11

Mechanics • Equilibrium in Three Dimensions

STEP II 2011, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q11

The key idea

Everything rests on writing a unit vector along each string. Each one splits into a vertical part, fixed by the string's angle with the vertical, and a horizontal part, which points from O towards the string's end in the ceiling and is fixed by the angle that direction makes with OA. Once the three unit vectors are known, equilibrium is just three scalar equations, one for each of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$.

Solution

We model P as a particle and the strings as light and inextensible, so each string pulls P towards its fixed end with a force along the string. Take O as the origin, with $\mathbf{j}$ along OA and $\mathbf{k}$ vertically upwards; $\mathbf{i}$ is horizontal and perpendicular to OA.

First the trigonometry. Since $\tan\theta = \sqrt{2}$, a right-angled triangle with sides $1, \sqrt{2}, \sqrt{3}$ gives

$$\sin\theta = \frac{\sqrt{2}}{\sqrt{3}}, \qquad \cos\theta = \frac{1}{\sqrt{3}}.$$

Since $\tan\phi = \frac{1}{4}\sqrt{2}$, the sides are $4, \sqrt{2}$ and hypotenuse $\sqrt{18} = 3\sqrt{2}$, so

$$\sin\phi = \frac{1}{3}, \qquad \cos\phi = \frac{2\sqrt{2}}{3}.$$

Part (i)

A string making angle α with the vertical has unit vector (from P towards the ceiling) with vertical component $\cos\alpha$ and horizontal part of length $\sin\alpha$.

For PB, $\alpha = 90^\circ - \theta$, so the vertical component is $\cos(90^\circ - \theta) = \sin\theta = \frac{\sqrt{2}}{\sqrt{3}}$ and the horizontal part has length $\sin(90^\circ - \theta) = \cos\theta = \frac{1}{\sqrt{3}}$.

The horizontal part points along OB (since O is directly above P), which makes angle $90^\circ + \theta$ with $\mathbf{j}$. A horizontal unit vector at angle $90^\circ + \theta$ to $\mathbf{j}$ is $\pm\sin(90^\circ + \theta)\,\mathbf{i} + \cos(90^\circ + \theta)\,\mathbf{j} = \pm\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}$. We choose the sense of $\mathbf{i}$ so that B lies on the

negative-**i** side of OA; the required form fixes this choice. The horizontal part of the unit vector along PB is then

$$\tfrac{1}{\sqrt{3}}(-\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}) = \tfrac{1}{\sqrt{3}}\left(-\tfrac{1}{\sqrt{3}}\,\mathbf{i} - \tfrac{\sqrt{2}}{\sqrt{3}}\,\mathbf{j}\right) = -\tfrac{1}{3}\,\mathbf{i} - \tfrac{\sqrt{2}}{3}\,\mathbf{j}.$$

Adding the vertical component, the unit vector in the direction PB is

$$-\tfrac{1}{3}\,\mathbf{i} - \tfrac{\sqrt{2}}{3}\,\mathbf{j} + \tfrac{\sqrt{2}}{\sqrt{3}}\,\mathbf{k},$$

as required. (Check: $\frac{1}{9} + \frac{2}{9} + \frac{2}{3} = 1$.)

Part (ii)

The same recipe gives the other two strings.

String PA: angle 30° with the vertical, horizontal part along **j**:

$$\mathbf{e}_A = \tfrac{1}{2}\,\mathbf{j} + \tfrac{\sqrt{3}}{2}\,\mathbf{k}.$$

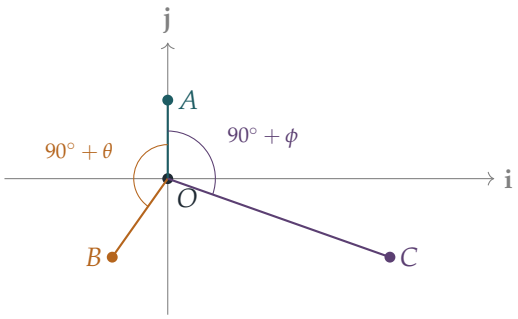
String PC: angle 60° with the vertical, so vertical component $\frac{1}{2}$ and horizontal length $\frac{\sqrt{3}}{2}$. The direction OC makes angle $90° + \phi$ with **j**, so its horizontal unit vector is $\pm\cos\phi\,\mathbf{i} - \sin\phi\,\mathbf{j}$.

Ask yourself Is C on the same side of OA as B, or the opposite side?

The diagram shows the opposite side, and equilibrium demands it anyway. The tension in PA has no **i**-component and that in PB has a negative one. If PC also had a negative **i**-component, nothing could balance them. So the sign is $+$, and

$$\mathbf{e}_C = \tfrac{\sqrt{3}}{2}\left(\tfrac{2\sqrt{2}}{3}\,\mathbf{i} - \tfrac{1}{3}\,\mathbf{j}\right) + \tfrac{1}{2}\,\mathbf{k} = \tfrac{\sqrt{6}}{3}\,\mathbf{i} - \tfrac{\sqrt{3}}{6}\,\mathbf{j} + \tfrac{1}{2}\,\mathbf{k}.$$

The plan view of the ceiling below (taking $OP = 1$, so $OA = \tan 30°$, $OB = \cot\theta$, $OC = \tan 60°$) shows the layout.



The forces on P are therefore

$$\begin{aligned} \text{tension in } PA: &\quad T\left(\tfrac{1}{2}\,\mathbf{j} + \tfrac{\sqrt{3}}{2}\,\mathbf{k}\right), \\ \text{tension in } PB: &\quad U\left(-\tfrac{1}{3}\,\mathbf{i} - \tfrac{\sqrt{2}}{3}\,\mathbf{j} + \tfrac{\sqrt{2}}{\sqrt{3}}\,\mathbf{k}\right), \\ \text{tension in } PC: &\quad V\left(\tfrac{\sqrt{6}}{3}\,\mathbf{i} - \tfrac{\sqrt{3}}{6}\,\mathbf{j} + \tfrac{1}{2}\,\mathbf{k}\right), \\ \text{weight}: &\quad -W\,\mathbf{k}. \end{aligned}$$

Part (iii)

P is in equilibrium, so the four forces sum to zero. We take components.

i:

$$-\tfrac{1}{3}U + \tfrac{\sqrt{6}}{3}V = 0 \quad\Longrightarrow\quad U = \sqrt{6}\,V,$$

as required.

j:

$$\tfrac{1}{2}T - \tfrac{\sqrt{2}}{3}U - \tfrac{\sqrt{3}}{6}V = 0.$$

Substituting $U = \sqrt{6}V$ and using $\sqrt{2}\sqrt{6} = 2\sqrt{3}$:

$$\tfrac{1}{2}T = \tfrac{2\sqrt{3}}{3}V + \tfrac{\sqrt{3}}{6}V = \tfrac{5\sqrt{3}}{6}V \quad\Longrightarrow\quad T = \tfrac{5\sqrt{3}}{3}V.$$

k:

$$\tfrac{\sqrt{3}}{2}T + \tfrac{\sqrt{2}}{\sqrt{3}}U + \tfrac{1}{2}V = W.$$

Substituting both results,

$$\tfrac{\sqrt{3}}{2}\cdot\tfrac{5\sqrt{3}}{3}V + \tfrac{\sqrt{2}\sqrt{6}}{\sqrt{3}}V + \tfrac{1}{2}V = \tfrac{5}{2}V + 2V + \tfrac{1}{2}V = 5V = W.$$

Hence

$$V = \frac{W}{5}, \qquad U = \frac{\sqrt{6}\,W}{5}, \qquad T = \frac{5\sqrt{3}}{3}\cdot\frac{W}{5} = \frac{W}{\sqrt{3}} = \frac{\sqrt{3}\,W}{3}.$$

All three are positive, so every string is taut, which is consistent with the model.

Checks

- Each unit vector has length 1: for $\mathbf{e}_C$, $\frac{6}{9} + \frac{3}{36} + \frac{1}{4} = 1$.
- The angle between the horizontal parts of $\mathbf{e}_A$ and $\mathbf{e}_C$ has cosine $-\frac{1}{3} = -\sin\phi$, so it is $90^\circ + \phi \approx 109.47^\circ$, as it should be.
- With $W = 1$: $T \approx 0.5774$, $U \approx 0.4899$, $V = 0.2$. Then the vertical components $0.5774 \times 0.8660 + 0.4899 \times 0.8165 + 0.2 \times 0.5 = 0.5 + 0.4 + 0.1 = 1$ balance the weight. Note the neat split: PA, PB, PC carry $\frac{1}{2}$, $\frac{2}{5}$, $\frac{1}{10}$ of the weight.

Common slips

- Mixing up sin and cos for the string PB: its angle with the vertical is $90^\circ - \theta$, so its vertical component is $\sin\theta$, not $\cos\theta$.
- Putting C on the same side of OA as B. The **i**-equation then forces a negative tension (that is, $V = -\frac{1}{3}W$), which a string cannot have.
- Forgetting that $\cos(90^\circ + \phi) = -\sin\phi$, which gives the wrong sign for the **j**-components of $\mathbf{e}_B$ and $\mathbf{e}_C$.

Beyond STEP

Three strings in three independent directions make the system *statically determinate*: the equilibrium equations form a 3×3 linear system with a unique solution. With a fourth string the tensions are no longer fixed by statics alone, and engineers have to bring in the elasticity of the strings. That is the start of structural analysis and of the linear algebra of rank and null spaces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats forces as vectors in this way, and Strang, *Introduction to Linear Algebra*, explains when such systems have unique solutions.

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